

Information Retrieval and Text Mining (Code: KEN4153). 6 ECTS

Master Artificial Intelligence & Master Data Science for Decision Making

Study year: 2025-2026

Period: 5

Contact Information

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Course Description and Intended Learning Outcomes

Examiner: Prof dr. ir. J.C. Scholtes

Tutor(s): DACS PhD student(s).

Desired Prior Knowledge: Proficiency in Python.

Prerequisites: Google CoLAB pro subscription for the tutorials. However, we will try to setup an alternative cloud environment before the start of the lectures. In that case the Pro subscription is not needed. For tutorial 11, you will need an OpenAI API key to generate tokens for your chatbot. This will cost you Eur 1-5 for. You will need to fund this key, otherwise it does not work.

Description: This course is all about algorithms for Conversational Search

Traditional search (aka Information Retrieval) is about “Finding the needle in the haystack”. This course focusses on more complex problems: “How does the needle look like and where is the haystack?” using text-mining, topic modeling and data visualization techniques. Today, search is more about, “Having a guide handing you the needle and explaining you its significance and context” using conversational AI in combination with search engines and text-mining.

Building a full-text search engine may look trivial, but it is not! How do you search hundreds of billions of documents that can be located anywhere, with sub-second responds times? How do you find exactly what you are looking for without missing relevant information or having to plough through hundreds of irrelevant documents? How can you find if you do not know exactly what you are looking for? How can you find information which is deliberately hidden? How do you know that your search engine has given you the right information? Where does it come from? Is the answer factually correct?

In this course, we will teach you how to address these questions in three steps: (1) how is a search engine is constructed, optimized and used effectively, (2) How can techniques from the world of text-mining, information extraction, text classification, clustering, topic modelling and data visualization lead to better search, and (3) What is the best way to integrate chatbots with search engines resulting in responsible conversational search. How to best guarantee factuality, measure and prevent hallucinations, provide provenance and explainability of the chatbots’ recommendations. How can we integration knowledge graphs and retrieval augmented generation (RAG) in the conversation? How can agentic architectures help us to overcome the limitations of LLMs?

Interestingly, traditional search-engine and text-mining techniques are essential components of such responsible conversational search solutions. In this course we will discuss all components and how they best work together.

Linear Algebra, Statistics, Deep Learning and Natural Language Processing play an important role in this course.

This course is complementary to the course Advanced Natural Language Processing (ANLP). Overlap is reduced to the necessary minimum. Both courses can be followed in any particular order. In the Information Retrieval and Text Mining course we focus more on creating an optimal modern search experience, in the Advanced Natural Language Processing course, we do a deep dive into the algorithms and models used for different language-related problems such as machine translation, abstracting, dialogs with chatbots, and LLM-based agents. Tutorials are shared between the two courses.

Prerequisites

Proficient coding skills in Python are required to participate in the tutorials.

Google CoLAB pro subscription for the tutorials. However, we will try to setup an alternative cloud environment before the start of the lectures. In that case the Pro subscription is not needed.

Knowledge and understanding: The student will be able to select, understand and apply different phases and methods used to create applications that exhibit an optimal search experience or provide excellent analytical insights for natural language. In addition, the student learns to evaluate the quality of such methods according to best-practice standards as used in the field.

Applying knowledge and understanding: Students will be able to recognize applications of text mining, information retrieval and conversational AI in different domains such as consumer search, legal services, medical research, regulatory oversight, compliance, digital humanities, and customer services. After the course, the student can formulate an opinion or course of action when dealing with text-based AI-problems based on incomplete, limited and in part unreliable information. After the course, students can apply their knowledge and understanding in a manner that shows a scientific approach to their work or vocation. They are able to handle complex and ill-defined text-based problems for which it is not a priori known if there is an appropriate solution, they know how to acquire the necessary information to solve the sub-problems involved, and they know how to proceed with problems for which there is no standard or reliable route to the solution.

Making judgements: Upon completion of the course, students are able to recommend the most appropriate methods from the fields of text mining, information retrieval and conversational AI when confronted with problems involving search and analysis of textual unstructured data.

Communication: Students are able to communicate the (dis)advantages of several methods from the field of text mining and information retrieval to both an audience of non-experts.

Learning skills: After the course, the student has developed those learning skills that are necessary for a successful further career in text mining or information retrieval at the highest professional level. The student will be able to continue to develop their text-mining and information retrieval skills. The student is able to detect missing knowledge and abilities and to deal with them appropriately by finding and consulting resources that can help them to fill the gaps and new developments.

Course Materials

- Study material: A syllabus with additional information on the course, the tutorials and copies of the course slides will be used. Both can be found on Canvas.

Recommended literature (not mandatory): Introduction to Information Retrieval. Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze. Cambridge University Press, 2008. In bookstore and online: <http://informationretrieval.org> and Feldman, R., and Sanger, J. (2006). The Text Mining Handbook: Advanced Approaches in Analyzing Unstructured Data. Cambridge University Press.

Study Load

An overview of the study load can be found here:

Estimated Workload	Information Retrieval & Text-mining		ECTS	6
			Hours	168
	Budget	Number	Time each	Total
Class Time	Lectures	12	2	24
	Lectures and tutorials	4	3	12
	Exam	1	2	2
Self Study	Prepare & assess content lectures	16	1	16
	Work on non-mandatory tutorials	1	24	24
	Work on mandatory tutorials	3	22	66
	Prepare exam	1	24	24
	Total			168

Assignments & Homework

Tutorials

We have developed these tutorials in Python to “experience” the theory yourself.

Some of these (the shorter ones) will be directly after the discussion of the theory.

Some are dedicated (2 hour) hands-on lectures.

You will have to do the tutorial individually.

There are several voluntary (example) tutorials to show you how the libraries we use work. There are 3 mandatory tutorials. Per tutorial, you will have to do several small tasks (code and open questions).

This course is about Information Retrieval, Text Mining, and Conversational Search — not about programming skills. The code cells in the tutorials show you *how* various methods work in practice using Python libraries. Focus on understanding the *concepts* and *results*.

Each tutorial also has several exercises at the end. These are typical examples of exam questions.

The mandatory tutorials are:

1. **03_IRTM_Search_Engines_2025_2026.ipynb**: build your own search engine, apply various relevance ranking schemes, and then create your own search engine on the text your favorite book or series.
2. **07_IRTM_Building_Structured_Representations_Part_2_2025_2026.ipynb**: use text mining techniques to create RAG vectors, KG, atomic facts, and Q&A for evaluation of the quality of your search engine and later on your chatbot.
3. **11_IRTM_Conversational_Search_Sticking_to_the_facts_2025_2026.ipynb**: create your own chatbot and apply various RAG techniques (and measure the quality quantitatively).

We will try to migrate the tutorials to our own cloud environment before the course starts. If we do not succeed, you can run the tutorials on your own computer. For the deep learning tasks you do need a GPU.

We expect that you submit the results for the mandatory tutorials on Canvas and in which you show you have executed the tutorial and you answer the additional questions and reflect on the results of the tutorial.

The file name of your submission should contain:

Your name

Your student number

Tutorial name

Finishing the tasks of the mandatory tutorials and submitting a copy of your results page is mandatory. The voluntary tutorials are highly recommended as these will make you understand the course material better and also show you examples of exam questions.

We highly recommend to follow the tutorials at the assigned times. We cannot guarantee that there is either a teaching assistant or a professor available for your questions on other times. This is related to the large number of students in this (popular) course.

The submission of the tutorials is due Wednesday 5pm of the week after the tutorials are given (also if you missed the lesson or tutorial) in the assignments section. If you are late, you get a penalty.

Examination

Each of the 3 (non-rounded) tutorial grades will count for 10% of your grade, so 30% in total. The non-rounded grade for the final exam will count as 70%. The result of this calculation will be rounded.

Tutorials have to be done individually. Please do not generate the answers with GPT, Gemini or another form of Generative AI. If you do so, you will most likely fail at the final exam.

It is mandatory to invest a minimal reasonable effort for each tutorial and the project. It is up to the discretion of the teaching assistant and the professor to decide what a “reasonable effort is” in case of doubt. But count minimally on spending the time on the tutorial as indicated above.

We will only grade your exam if you finished the tutorials to our satisfaction.

You will minimally have to score a 5.5 for the final exam to pass this course! So, if you have a 5.49 for the exam and a 10 for the mandatory tutorials, you still fail.

The final exam is a **closed** book exam. We can also ask questions about the meaning of python code using several methods.

There will be a resit. For this the same requirements apply.

If you did the tutorials or project last year, these will not be submissible this year due to the many differences in this year's course compared to last year. You have to redo the mandatory 3 tutorials. **Resit**

- A resit is available, but only if you participated in the mandatory tutorials from the start of the lectures.

Grading

- The tutorials will be graded based on the points scored for the tasks and questions. Different tasks and questions may have different weights per tutorial. For the final tutorial grade, we will calculate the average of your mandatory tutorial grades.
- The exam will be graded based on the points scored for the various questions. Different questions may have different weights.
- For the final grade, we will use the non-rounded results for the tutorials and the exam.

Please do not generate the answers with GPT, Gemini or another form of Generative AI. If you do so, you will most likely fail at the final exam.

Course Schedule (assignment deadlines)

Course Outline and Dates for Year 2024-2025

Lecture	Date and time	Description Contents of Course
1	Thursday April 9, 2026. 13.30-15.30	• Structure course and contents • Explanation Examination • Explanation Tutorials • Start Information Retrieval: search index, inverted files, and tokenization.
2	Thursday April 9, 2026. 16.00-18.00	• Information Retrieval. a. Full-text queries b. Optimization index & compression c. Incomplete searching d. Incremental indexing
3	Friday April 10, 2026. 11.00-13.00	• Relevance Ranking, Relevance Feedback & Recommender: a. TF-IDF b. Okapi BM25 c. Vector Space d. Page Rank e. Knowledge Graphs f. Neural Methods
4	Thursday April 16, 2026. 13.30-15.30	• Quantitative Validation Techniques • Tutorial

5a	Thursday April 10, 2026. 16.00-18.00	• Tutorial Search Engines Part 1.
5b	Friday April 11, 2026. 11.00-13.00	• Tutorial Search Engines Part 2.
6	Thursday April 23, 2026, 13.30-15.30	• Building Structured Representations from Text – Part 1 a. The Problem with Language b. Paragraph and Sentence Detection c. Syntax and Semantics d. Coreference & pronoun resolution e. Negation handling
7	Thursday April 23, 2026. 16.00-18.00	• Building Structured Representations from Text – Part 2 a. Named Entity Extraction b. Relation Extraction c. Entity normalization, linking & disambiguation d. Building Graphs from Language e. Searching Graphs & GNN
8	Friday April 24, 2026. 8.30-10.30	• Detecting Patterns and Organizing Text – Part 1 a. Text Classification b. Decision Trees c. Naïve Bayes d. Logistic Regression e. k-NN f. Support Vector Machines (SVM) g. Sentiment & Emotion Analysis h. Toxic Text Detection
9	Thursday April 30, 2026. 13.30-15.30	• Detecting Patterns and Organizing Text – Part 2 a. Clustering

		b. Topic Modeling c. Anomaly Detection Tutorial Topic Modeling
10	Thursday April 30, 2026. 16.00-18.00	Conversational Search – the Basics
10a	Friday May 1, 2026. 11.00-13.00	Tutorial Conversation Search
11	Thursday May 7, 2026. 13.30-15.30	Conversational Search – RAG and KG
12	Thursday May 7, 2026. 16.00-18.00	Conversational Search - Agentic
13	Friday May 8, 2026. 11.00-13.00.	Exam preparation and Q&A
EXAM	Week of May 18 th , 2026	Exams (exact date to be announced later by the DACS examination administration)

Information for Repeaters

As the tutorials are completely new, you will have to do them again.